

EES 301: Global biogeochemical cycles and Climate change

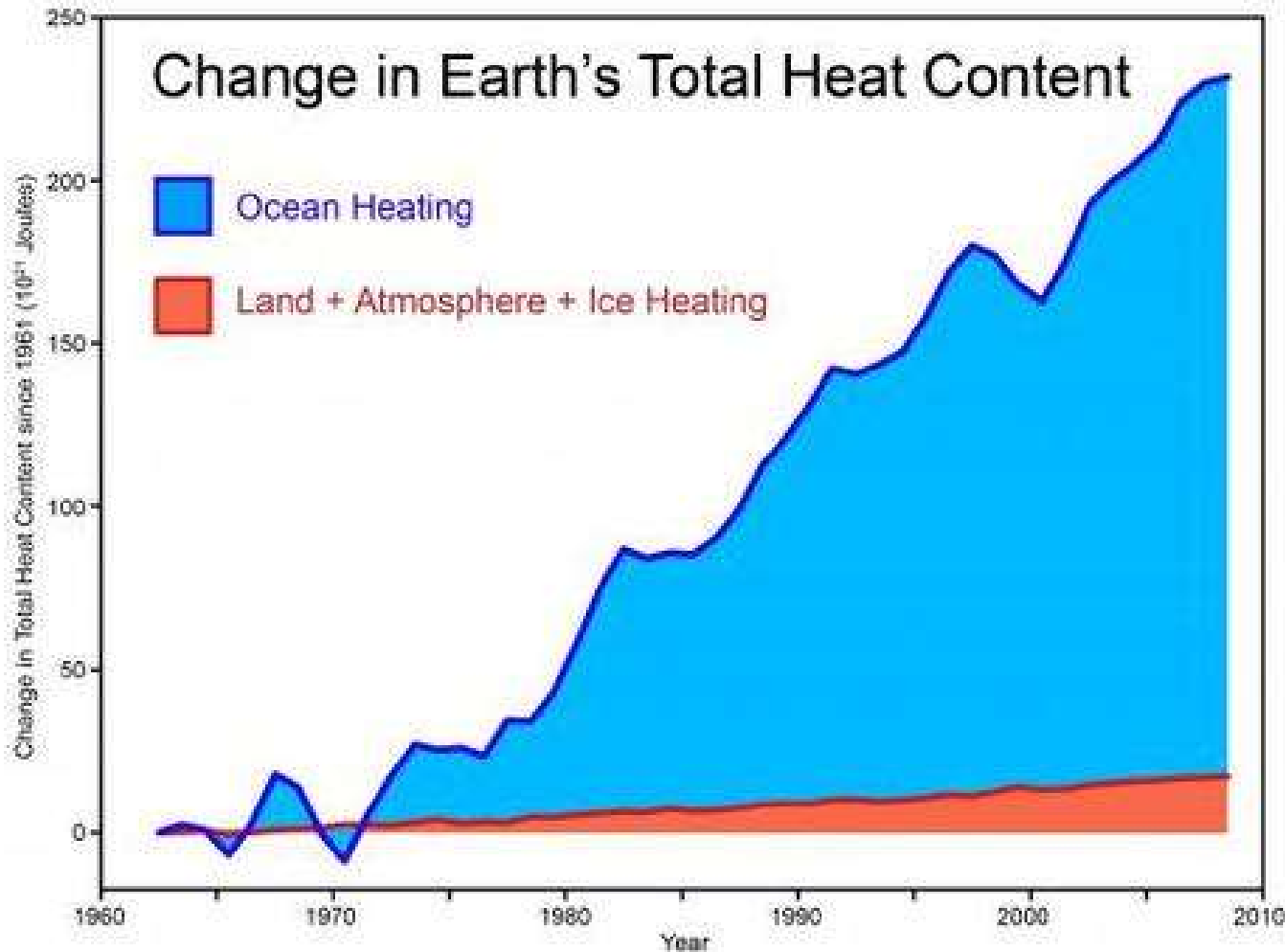


Radiative forcing the definition

- Radiative forcing is the difference between insolation (sunlight) absorbed by the Earth and energy radiated back to space.
- All changes to Earth's radiative equilibrium, that cause temperatures to rise or fall over decadal periods, are called climate forcings.
- Positive radiative forcing means Earth receives more incoming energy from sunlight than it radiates to space. This net gain of energy will cause warming.
- Conversely, negative radiative forcing means that Earth loses more energy to space than it receives from the sun, which produces cooling. A system in thermal equilibrium has zero radiative forcing.

Where has the additional energy gone

Anthropogenic GHG emissions have led to a substantial positive radioactive forcing



90% has gone into the ocean while 10% has been accommodated on the surface in the form of atmospheric heating, land heating and ice melting.

Over time this change in ocean energy will result in thermal expansion of ocean water and the ice melting adds additional water in the ocean. Both of these elements contribute to sea level rise.

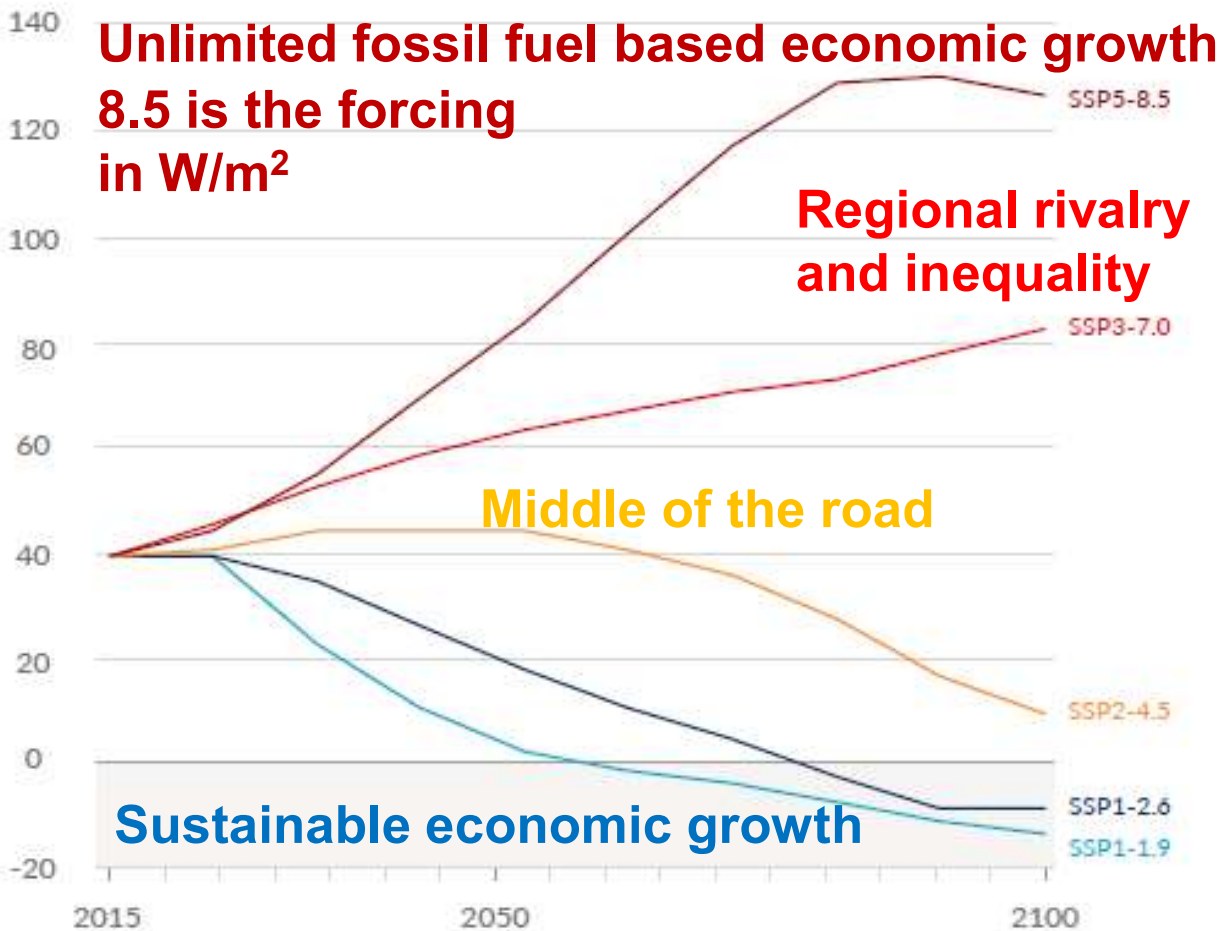
This heating will also reduce the amount of CO₂ the ocean can store

What climate change scenarios does the IPCC model?

Future emissions cause future additional warming, with total warming dominated by past and future CO₂ emissions

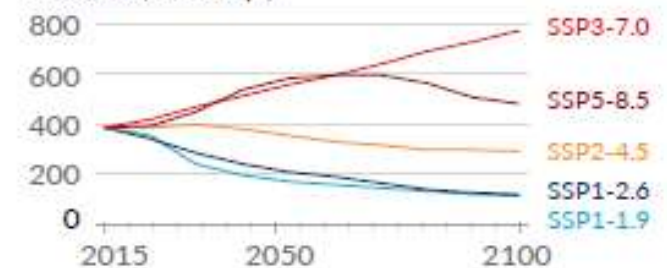
a) Future annual emissions of CO₂ (left) and of a subset of key non-CO₂ drivers (right), across five illustrative scenarios

Carbon dioxide (GtCO₂/yr)

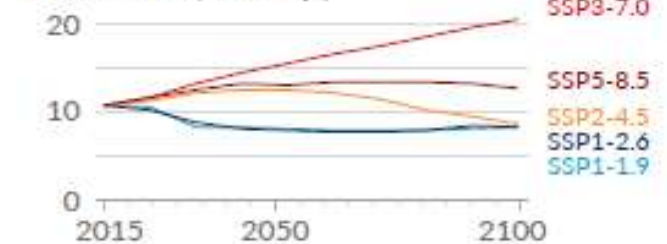


Selected contributors to non-CO₂ GHGs

Methane (MtCH₄/yr)

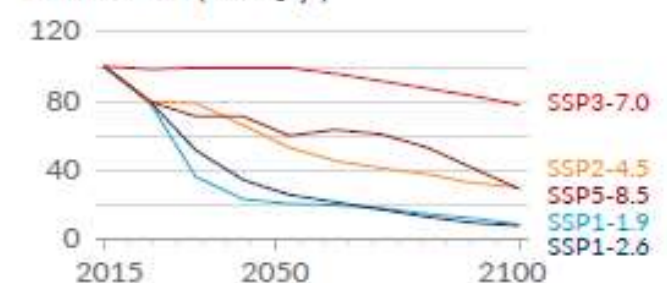


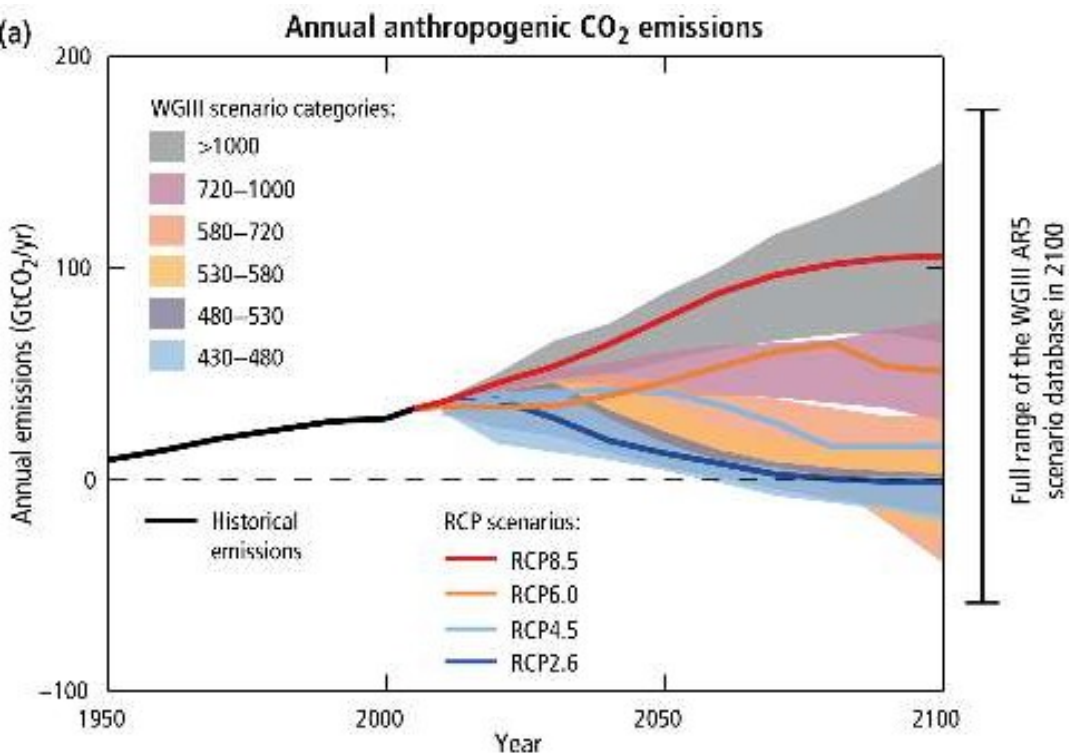
Nitrous oxide (MtN₂O/yr)



One air pollutant and contributor to aerosols

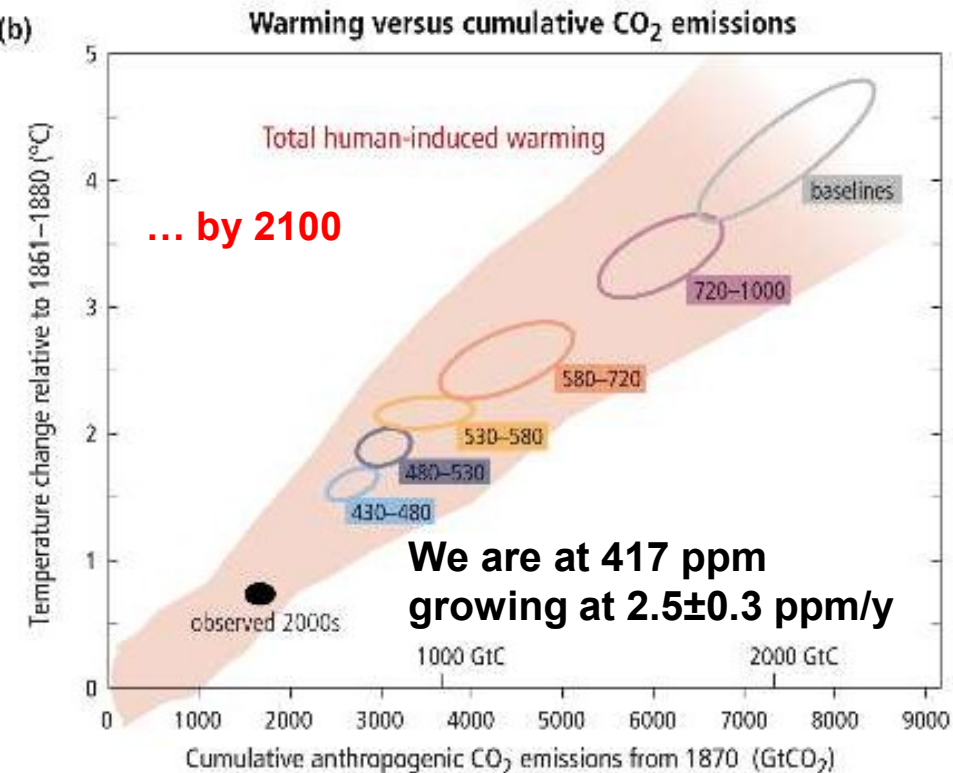
Sulfur dioxide (MtSO₂/yr)



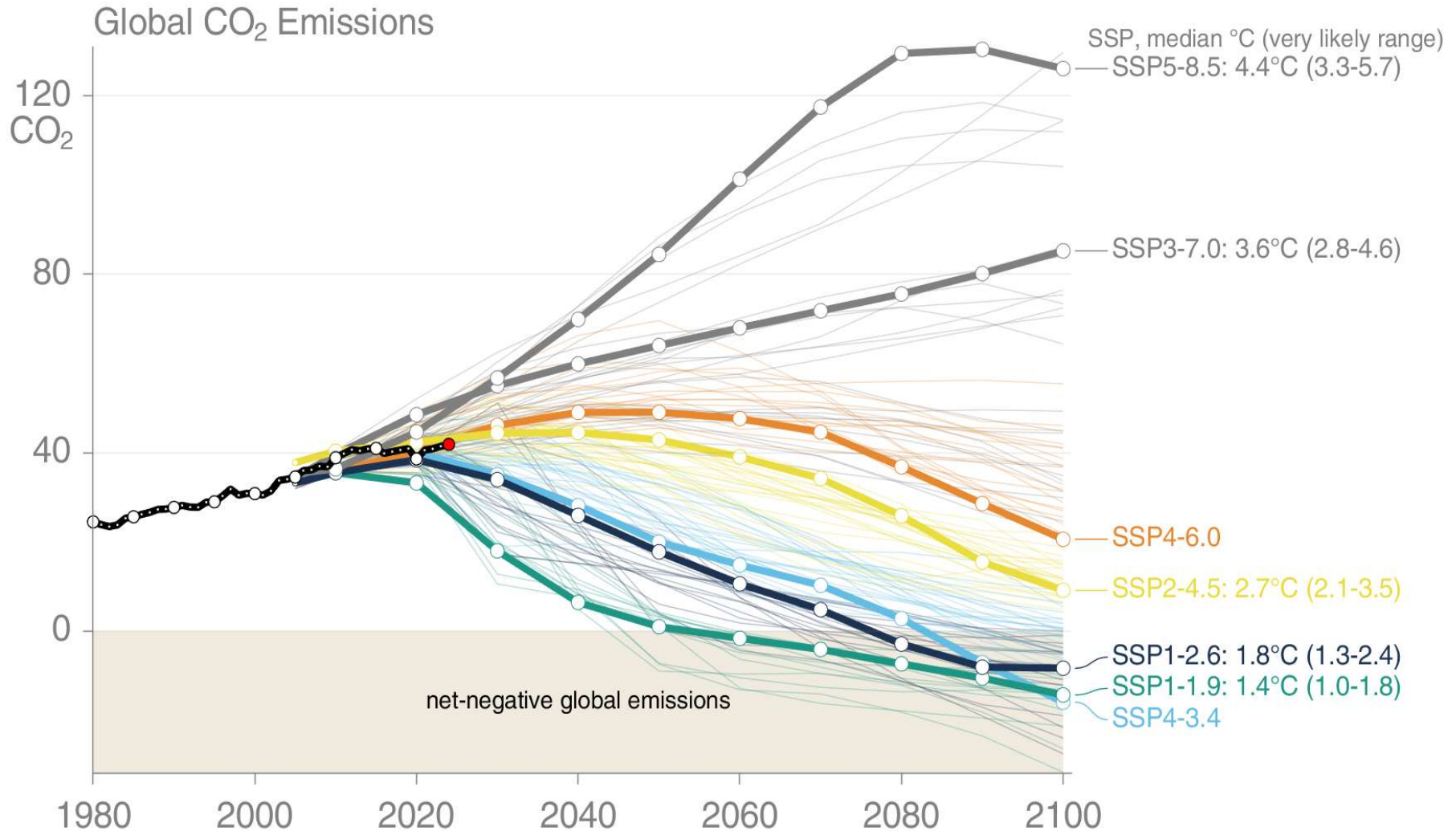


Climate change in the 5th assessment was modelled according to RCP “Representative Concentration Pathways” the number stands for the radiative forcing +2.6, +4.5, +6.0, and +8.5 W/m² relative to preindustrial values attained by 2100

- For RCP2.6 short term warming by 2100 will be +1.5 to +2 °C. Sustained net negative anthropogenic GHG emissions after the year 2070 are predicted. Global warming will reduce to 0.0 to 1.2 °C by the late-23rd century (2281-2300 average)
- For RCP8.5 short term warming by 2100 will be +4 to +5 °C. For the extended RCP8.5, global warming of 3.0 to 12.6 °C is projected by the late-23rd century
=> the uncertainty of the range stems from the fact that we do not know how fast the methane hydrates will melt. This frozen methane ice is on the ocean floor and its melting would catapult the planet into a “runaway greenhouse” which would take ~170,000 years to return to normal through natural processes



Which route are we on?



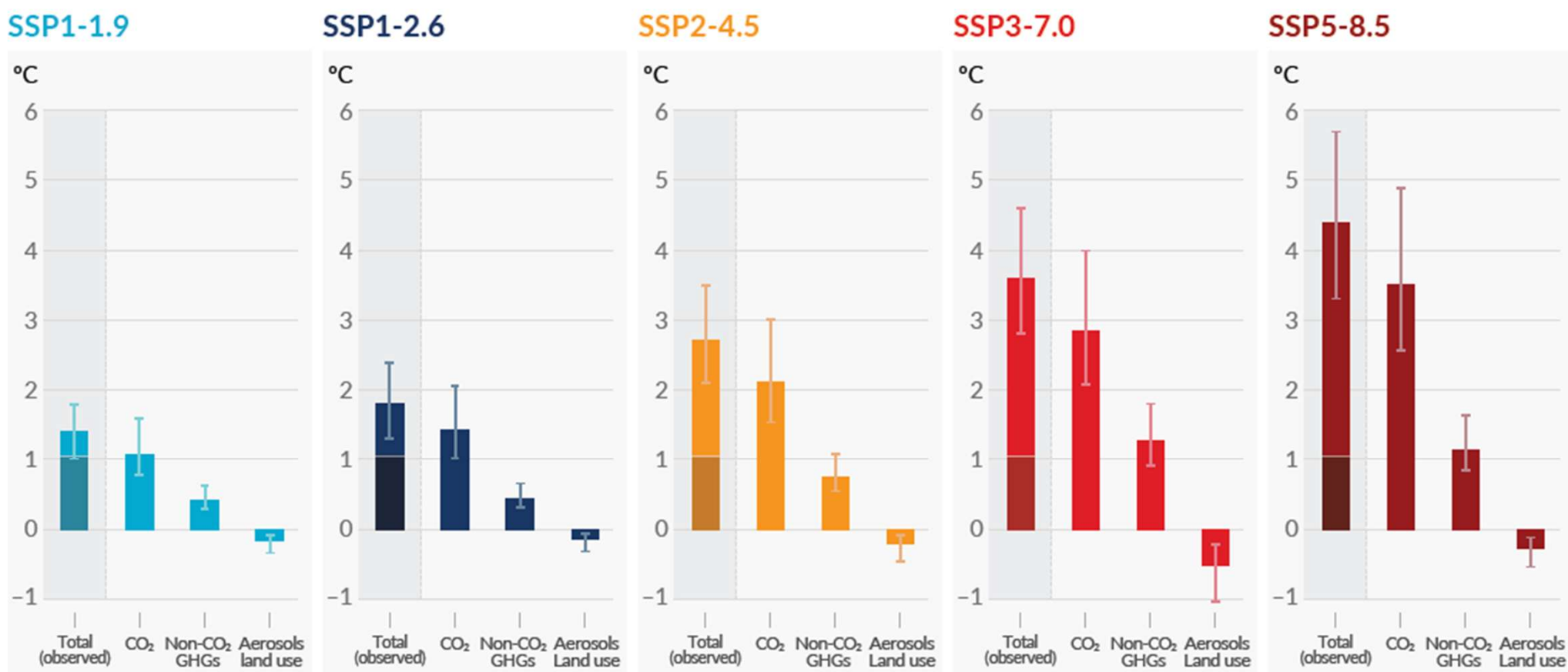
© Global Carbon Project • Data: Riahi et al (2017), Rogelj et al (2018), SSP Database (version 2)

<https://globalcarbonbudget.org/gcb-2024/>

Climate change in the 6th assessment had an additional “Representative Concentration Pathway” +1.9 W/m² relative to preindustrial values attained by 2100 which is compatible with the 1.5°C warming proposed for the Paris agreement

(b) Contribution to global surface temperature increase from different emissions, with a dominant role of CO₂ emissions

Change in global surface temperature in 2081–2100 relative to 1850–1900 (°C)



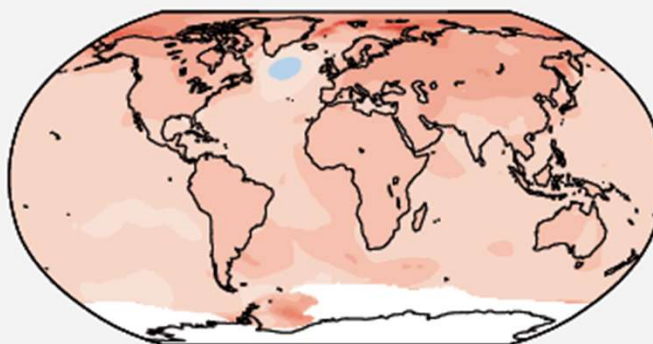
Total warming (observed warming to date in darker shade), warming from CO₂, warming from non-CO₂ GHGs and cooling from changes in aerosols and land use

Temperature responses will be largest in the Arctic where the ice albedo effect amplifies them

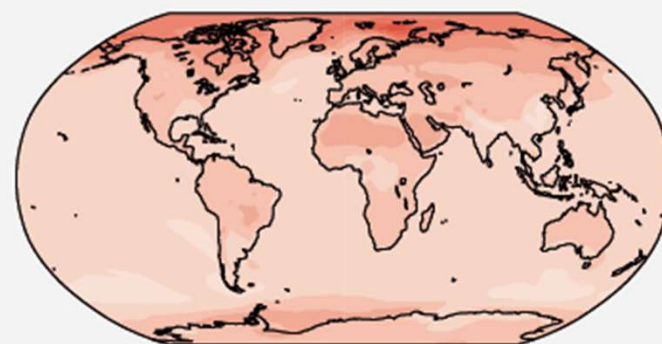
(a) Annual mean temperature change (°C) at 1°C global warming

Warming at 1°C affects all continents and is generally larger over land than over the oceans in both observations and models. Across most regions, observed and simulated patterns are consistent.

Observed change per 1°C global warming



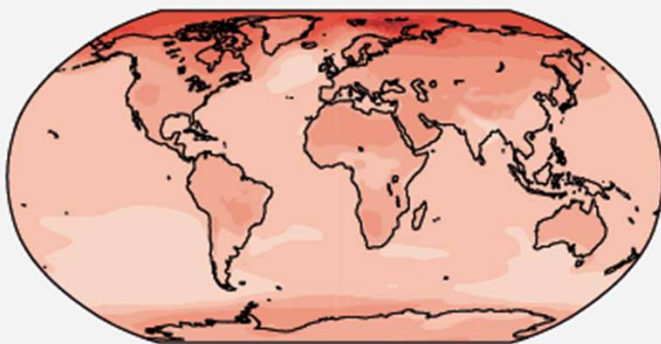
Simulated change at 1°C global warming



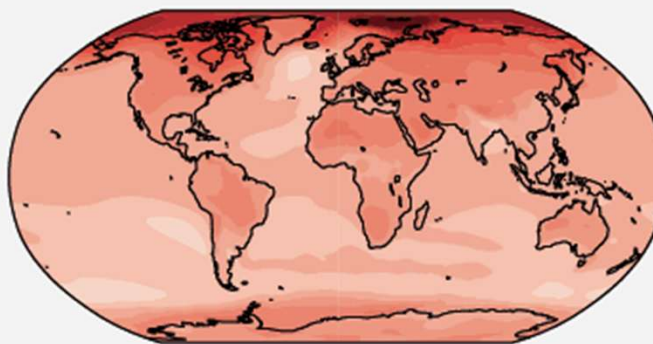
(b) Annual mean temperature change (°C) relative to 1850–1900

Across warming levels, land areas warm more than ocean areas, and the Arctic and Antarctica warm more than the tropics.

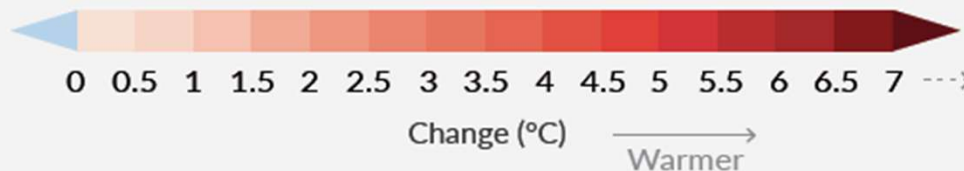
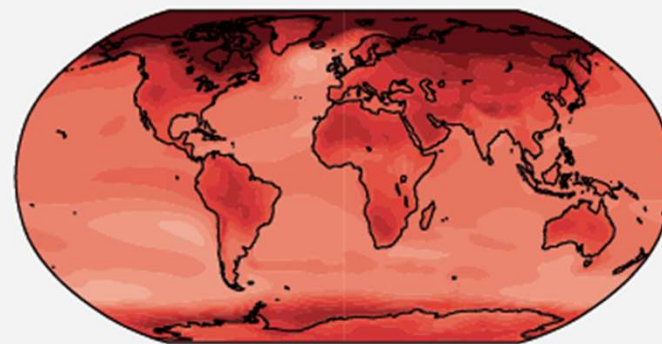
Simulated change at 1.5°C global warming



Simulated change at 2°C global warming



Simulated change at 4°C global warming

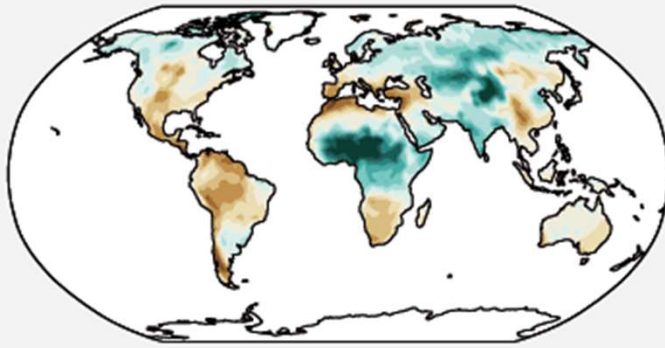


Biomes are expected to shift

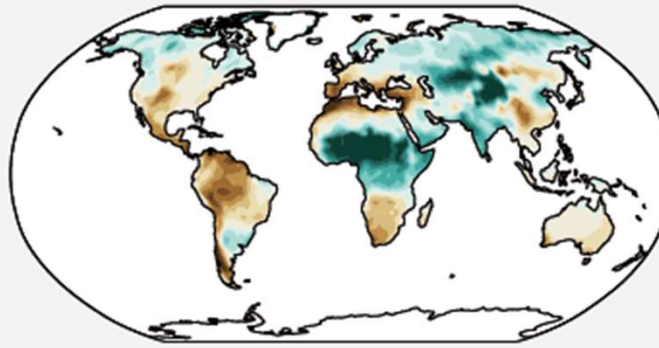
(d) Annual mean total column soil moisture change (standard deviation)

Across warming levels, changes in soil moisture largely follow changes in precipitation but also show some differences due to the influence of evapotranspiration.

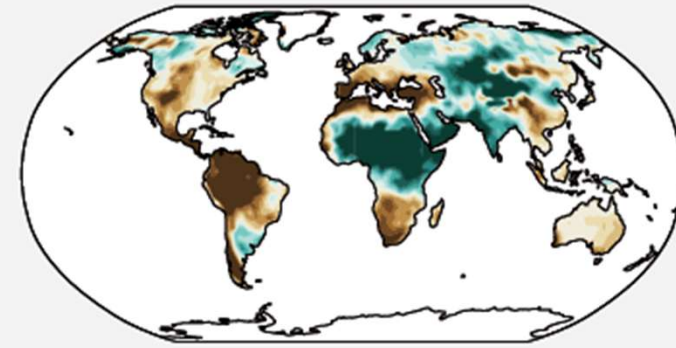
Simulated change at 1.5°C global warming



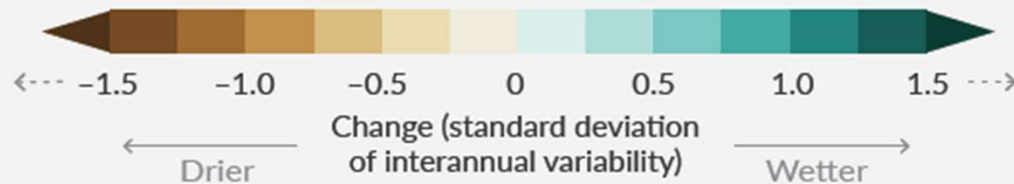
Simulated change at 2°C global warming



Simulated change at 4°C global warming



Relatively small absolute changes may appear large when expressed in units of standard deviation in dry regions with little interannual variability in baseline conditions.

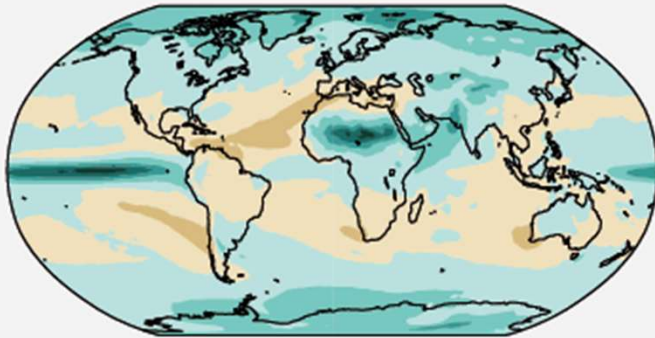


Rainfall extreme become more frequent

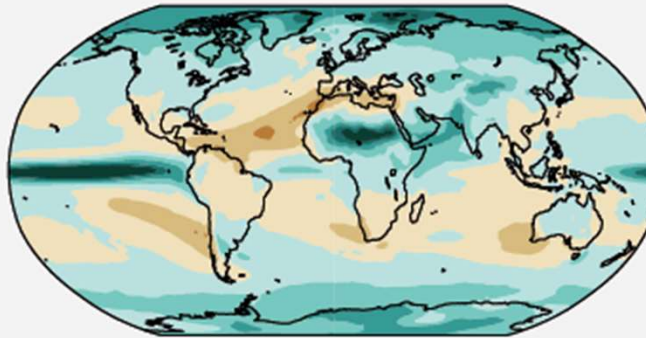
(c) Annual mean precipitation change (%) relative to 1850–1900

Precipitation is projected to increase over high latitudes, the equatorial Pacific and parts of the monsoon regions, but decrease over parts of the subtropics and in limited areas of the tropics.

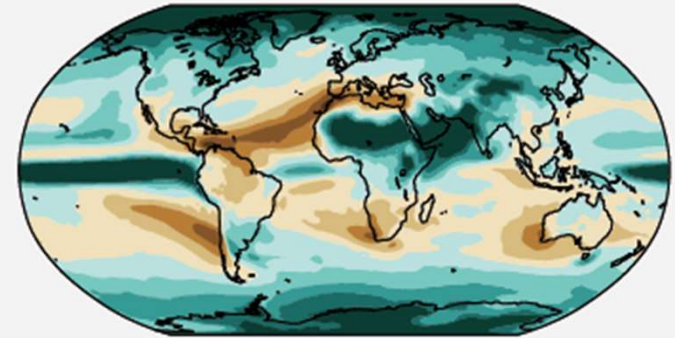
Simulated change at 1.5°C global warming



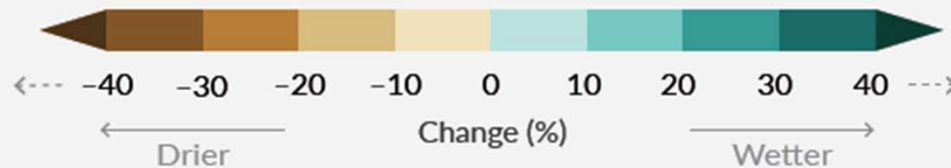
Simulated change at 2°C global warming



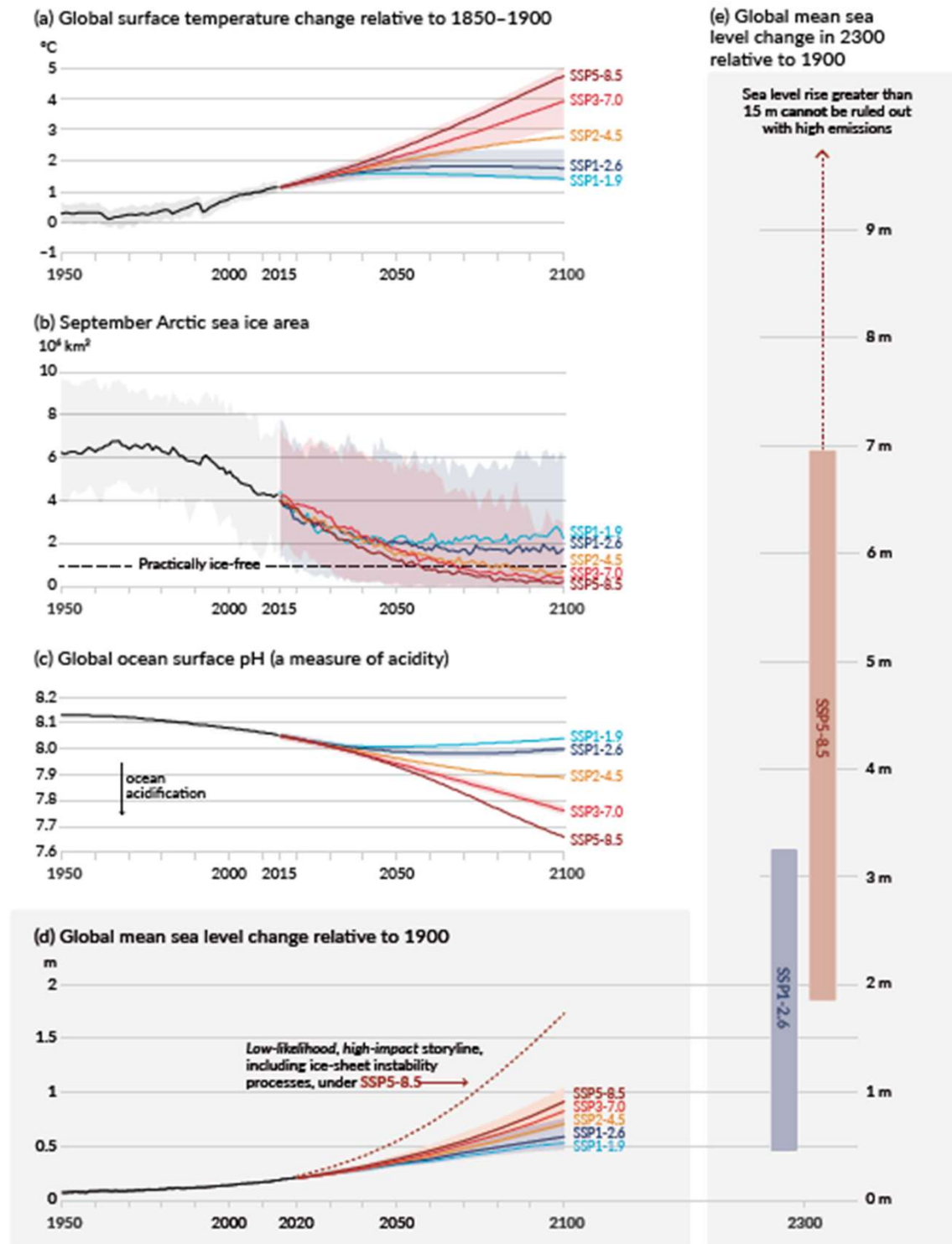
Simulated change at 4°C global warming



Relatively small absolute changes may appear as large % changes in regions with dry baseline conditions.



1. Arctic will become ice free in summer under all but the most benign scenarios
2. Ocean pH will drop under the most extreme scenarios hampering phytoplankton growth, CO₂ sequestration and O₂ production
3. Recently discovered ice sheet process are still only included as a dashed line meaning sea level rise could be more rapid than anticipated.
4. The message that sea levels will continue to rise beyond 2100 is finally included in the summary for policy makers



#ALLFORCLIMATE: SIX DEGREES

We are facing a climate emergency posing an existential threat to humanity. The crisis requires “rapid, far-reaching and unprecedented changes in all aspects of society”, says the UN’s IPCC*. If we want to ensure good living conditions for all, within the limits of the planet, we must redefine what living well means, and create an economy that does not depend on growth.

We have been waiting too long for others to solve the crisis. But all actors in society have important roles to play in the transition ahead. All of us, individuals and organisations, must realise that we are part of the problem and must act boldly now to become part of the solution. This is the core of the #AllForClimate campaign.

Despite the progress made by science, we do not know for sure when tipping points will be triggered, and when global warming could be unstoppable. For any tenth of a degree of global warming we let happen, we are gambling with a future that, in the worst case, none of us can survive.

** Intergovernmental Panel on Climate Change*



+0°C

THE WORLD BEFORE INDUSTRIALISATION

The temperature reached on Earth around 11,000 years ago after the last ice age. Ice melting made the sea level rise about 120 metres. Stable and mild climate allowed humans to grow food and settle down. Cities arose, consumption was low and living ways mostly sustainable. Humans were already affecting their environment, but not so much the world’s climate.

+1°C

EVERY TENTH OF DEGREE OF WARMING COUNTS

In 2015, global warming reached 1°C above pre-industrial levels. Six years later it reached 1.1°C. The Arctic and Antarctica are melting. The Gulf Stream is weakening. Glaciers are melting affecting freshwater supply. Hurricanes, droughts, heatwaves and large wildfires are more frequent. Low-lying coastlines are flooded. Ice-free sea absorbs more heat and accelerates global warming. Corals are bleaching. Wildlife is in decline. Food production is starting to be affected.

+2°C

A PRESCRIPTION FOR LONG-TERM DISASTER

Climate breakdown. Ice-free summertime in the Arctic. Irreversible loss of the West Antarctic Ice Sheet. 80 million people displaced due to sea-level rise. Deadly dengue fever. About half a million die of malnutrition. Insect crops pests. Hundreds of millions exposed to deadly heat. Shortage of water in Africa. Mass migration. Loss of glaciers affects freshwater supply. More floods. Conflagration of the Amazon rainforest. Many species are threatened. 99% of reef-building corals are lost.

+3°C

VEGETATION EMIT CARBON AND SPEED WARMING

Ice sheets and glaciers partly collapse. Sea-level rises to 8-14 metres. Extreme high tides, floods, superstorms and super-hurricanes. Hundreds of millions of people are forced to relocate. Half the world's population is exposed to deadly heat waves. Invading deserts. Food production cut by half. Mass starvation. Death of the Amazon rainforest. Wildlife disruption. Carbon release from vegetation and soils speeds global warming, so does methane emissions from melting permafrost.

+4°C

GLOBAL WARMING IS UNSTOPPABLE

Three-quarters of the global population is exposed to deadly heat in days. Substantial areas of the planet are biologically uninhabitable for humans. Climate refugee numbers become unmanageable. Half the world's land surface becomes 'arid'. Extreme droughts and floods. Category 6 super-hurricanes. Synchronous failures of major world food crops. Potential ice-free planetary state. Mass extinction. Permafrost meltdown makes the escalation of global warming unstoppable.

+5°C

SURVIVING HUMANS LIVE LIKE ANIMALS

Ice sheet is permanently gone. Humans can no longer survive outside for a long time. Food production is decimated. Complex human societies have collapsed. Global famine. Surviving humans try vainly to live like animals off the land.

+6°C

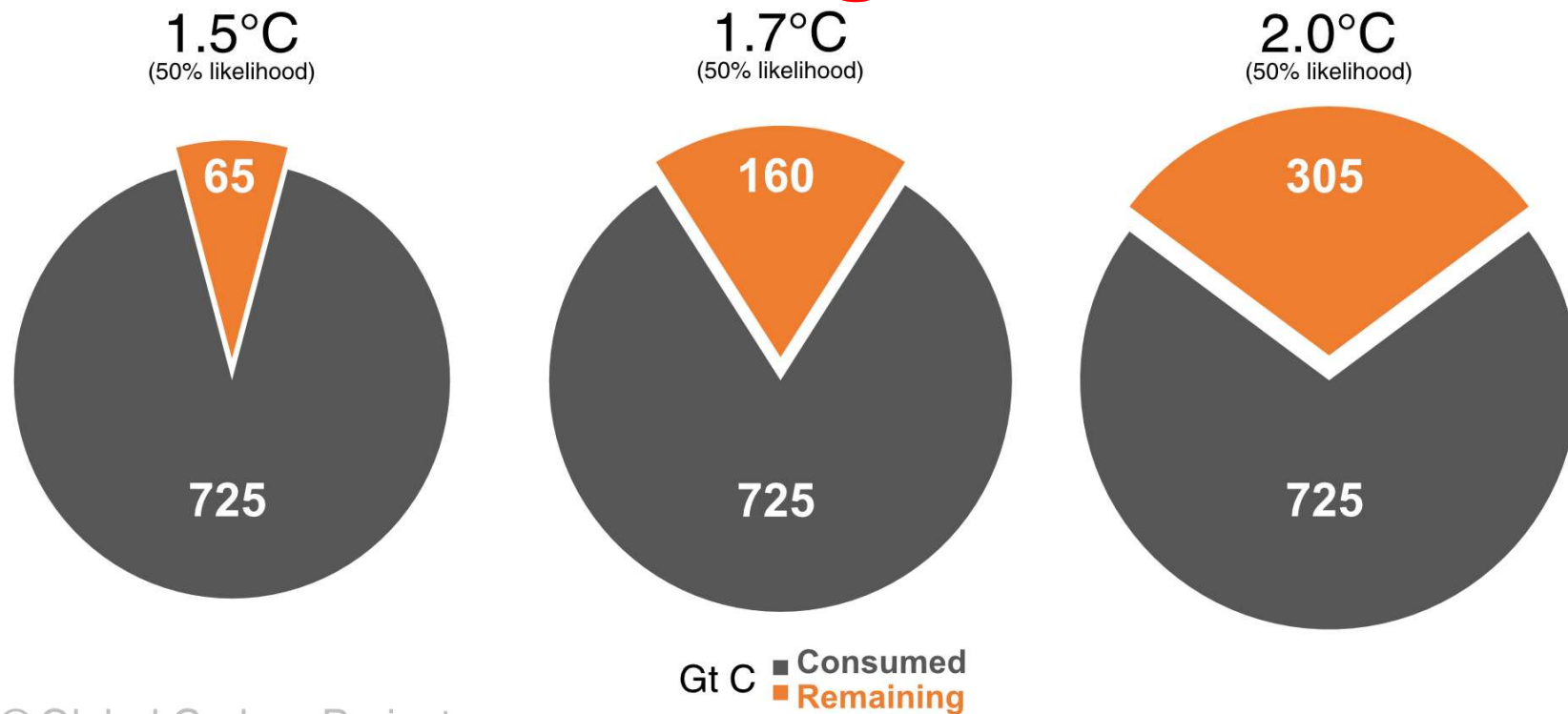
ONLY FEW ANIMALS AND FUNGI SURVIVE

Extreme greenhouse effect destroys most of life on Earth. Only a few animals can survive. Ecosystems and food webs cease to exist in any functional sense. The big winners are detritivores, bacteria and fungi.

The descriptions on this poster are mainly based on the book "Our Final warning: six degrees of climate emergency" written by award winning Journalist Mark Lynas and published in 2020.

Read more at www.allforclimate.today #ClimateForAll #AllForClimate 

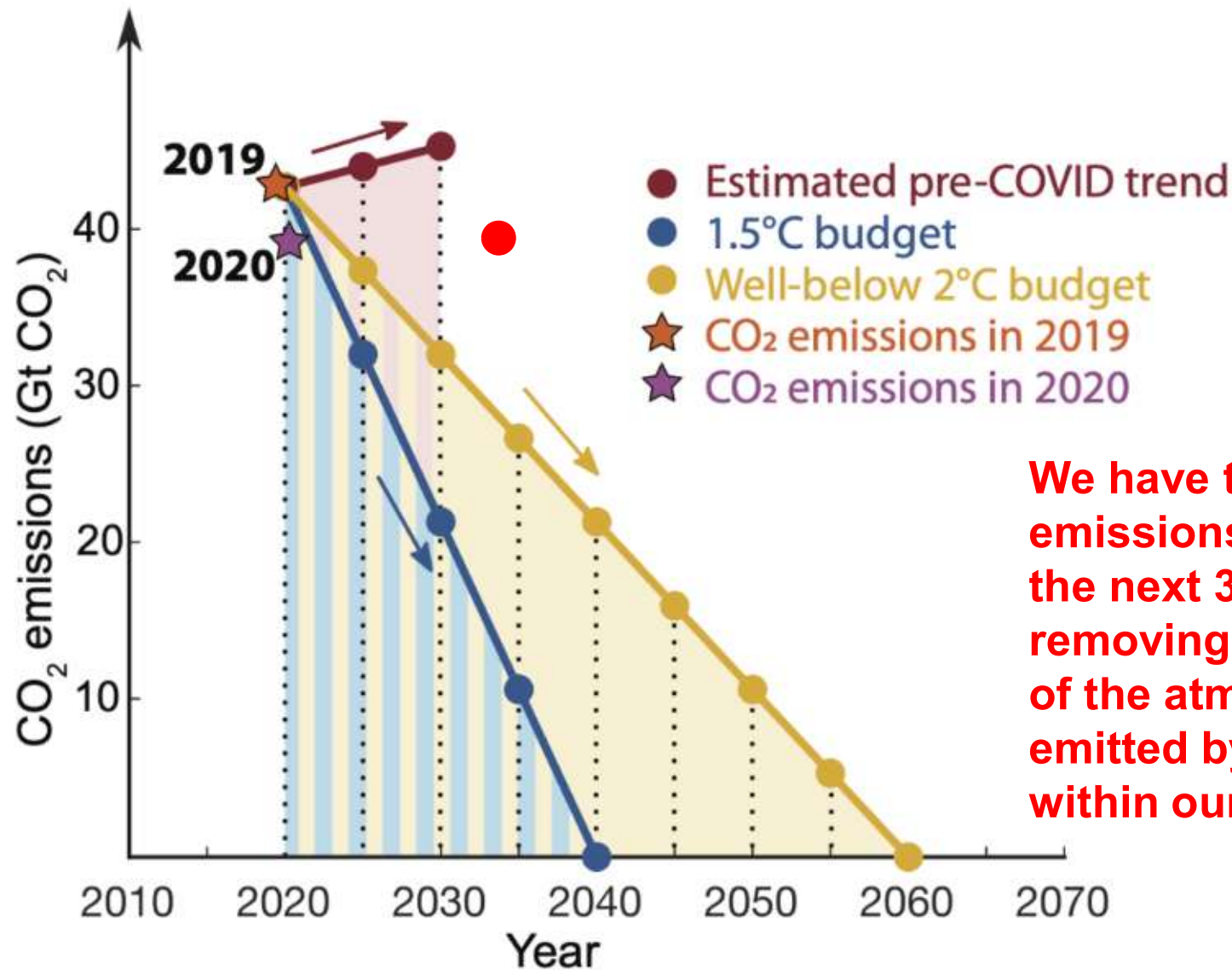
How urgent combatting climate change?



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- The budget is non-negotiable. It represents the safe zone for humanity.
- Paris agreements are not going to help. These are not good enough to keep us even within +2°C they will land us at +3°C
- We have less than 6 years of emissions at the current annual emission rate left for the +1.5°C budget. In fact the global average temperature last year was: +1.5°C

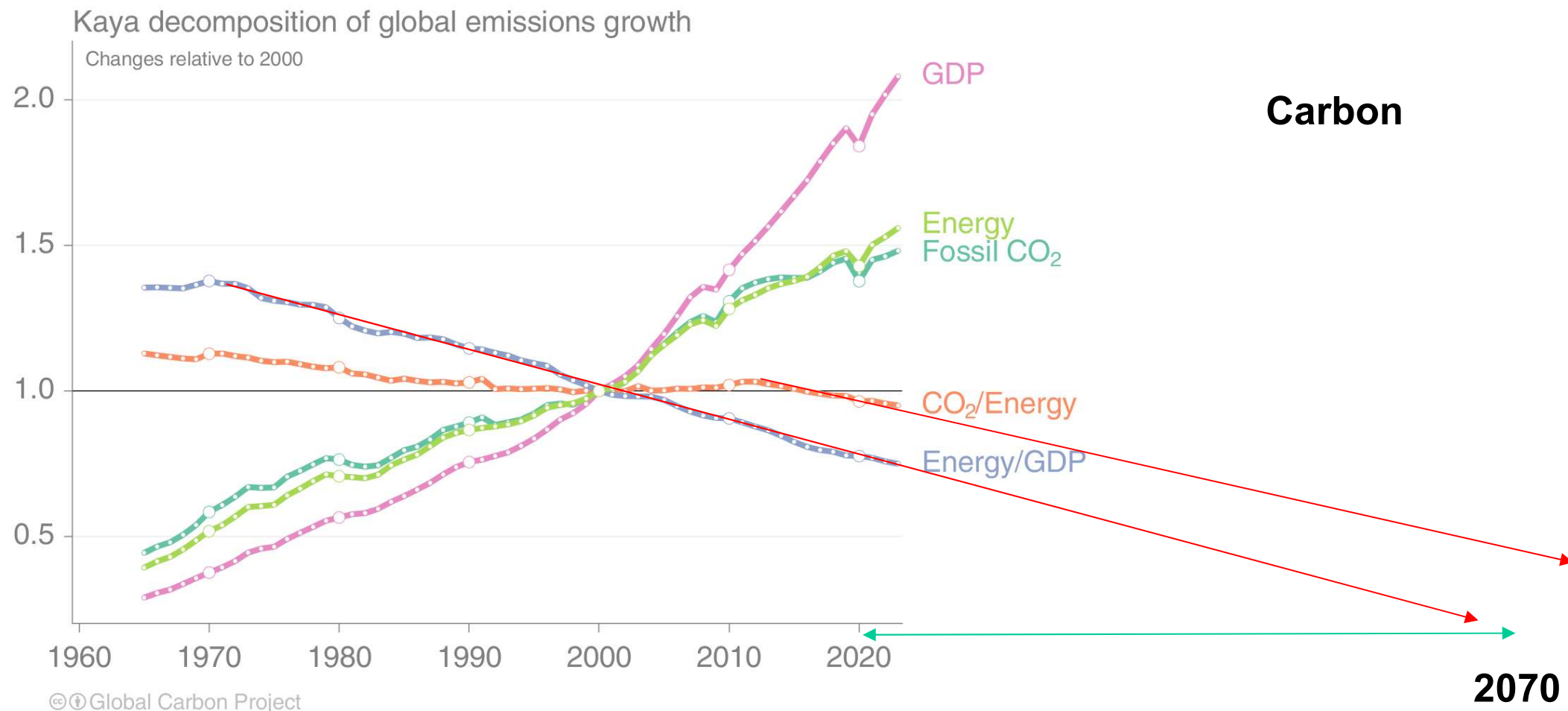
The ambitions gap



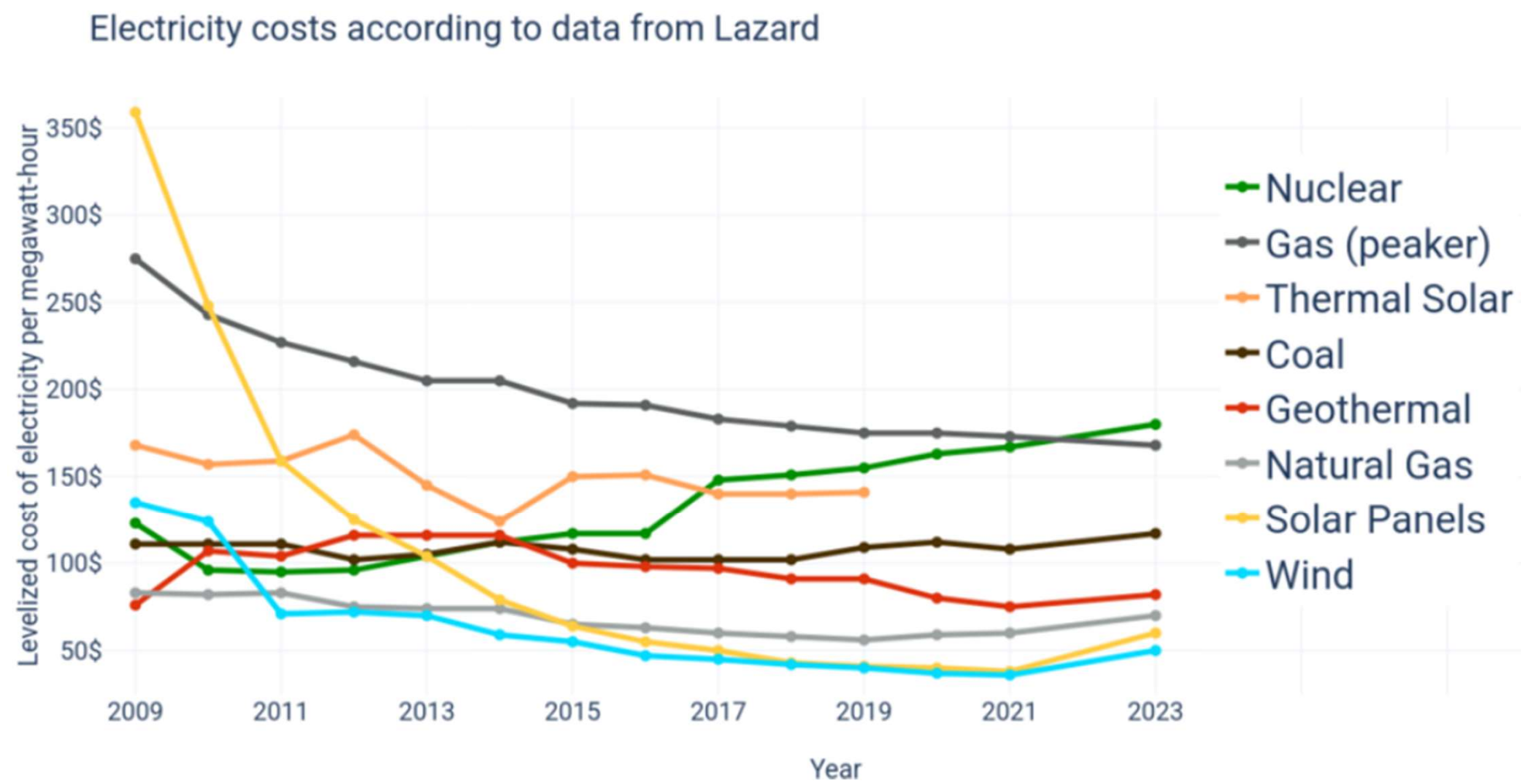
We have to reduce global emissions by 9% p.a. for the next 30 years and start removing more carbon out of the atmosphere than is emitted by 2050 to stay within our budget.

Anyone promising 0 kg CO₂/\$GDP by 2070 isn't promising you any special effort at all – they are merely promising their industry will survive price competition.
The cheapest energy is the one that you don't need in the first place.
Followed by renewable energy you produce yourself.

The Kaya decomposition illustrates that relative decoupling of economic growth from CO₂ emissions is driven by improved energy intensity (Energy/GDP) and, recently, carbon intensity of energy (CO₂/Energy)



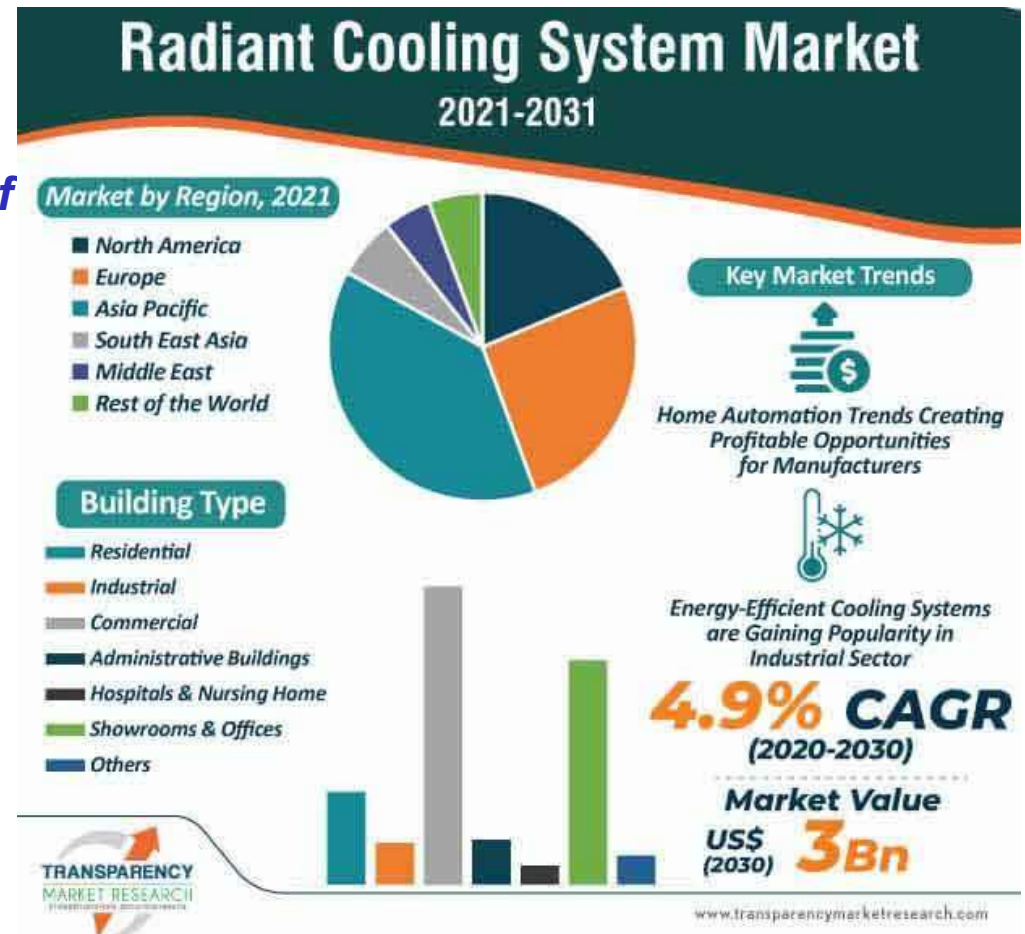
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Energy efficiency is king & cooling the key

- Right now cooling applications particularly of large data centres consume most energy worldwide
- The atmospheric window is the potential magic wand of cooling... materials that emit IR in that wavelength range while reflecting all incoming solar radiation can use space as cold sink! Cooling meets geo-engineering!

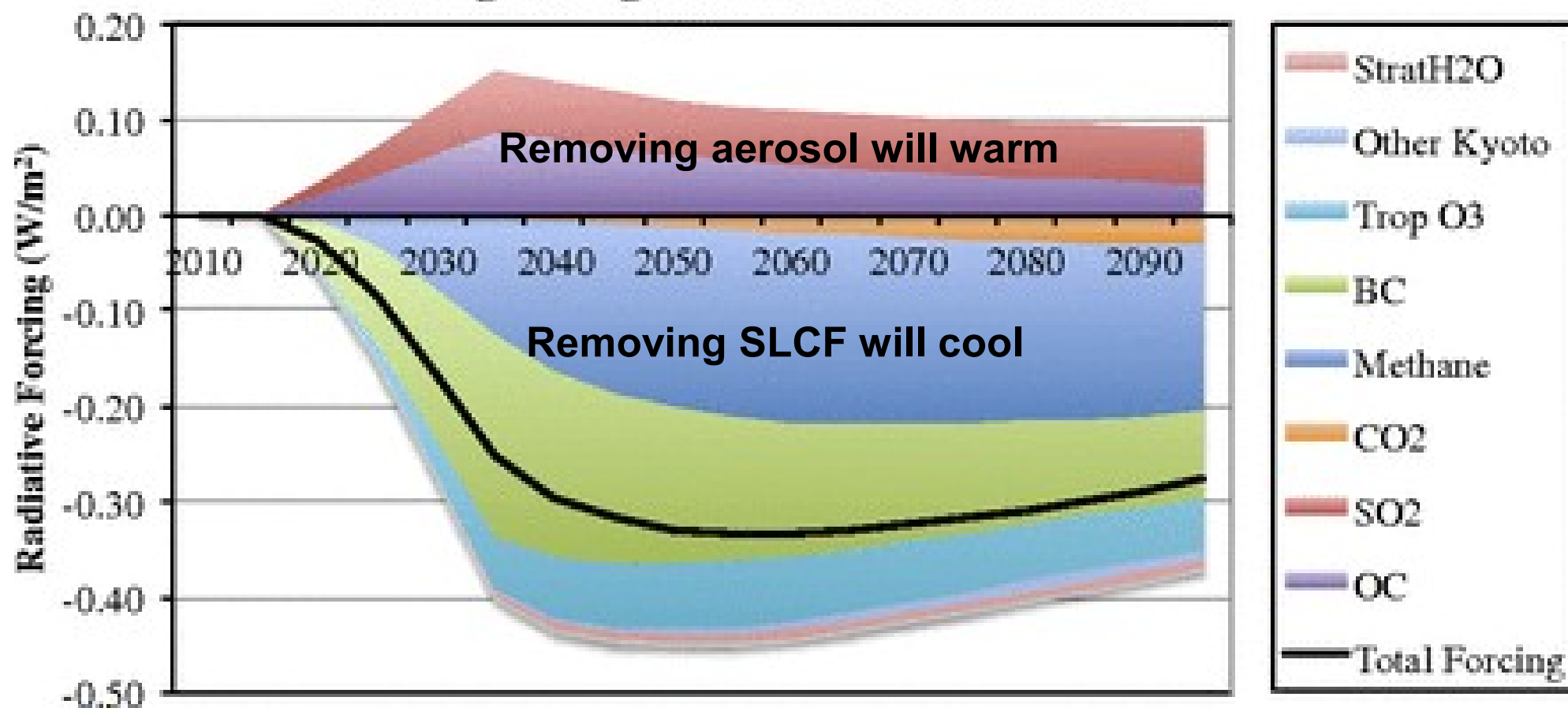
Bhatia et al. 2018 *Nature Communications* 9: 5001 “Measurements performed around solar noon show a minimum temperature of 6 °C below ambient temperature and maximum cooling power of 45 W m⁻². Our passive cooling approach, realized using commonly available low-cost materials, could improve the performance of existing cooling systems and enable next-generation thermal management and refrigeration solutions.” (This is an out of the box approach using cheap materials & a solar shield with a tracking device)



Mitigation of gases with a short lifetime can buy us most time

- Tropospheric ozone has a radiative forcing of $\sim 0.35 \text{ W/m}^2$ and is continuously produced and destroyed.
- If our transport sector was shifted to EV and energy generation was completely shifted to renewables, tropospheric ozone mixing ratios would return to preindustrial levels rapidly
- In comparison: even if we could stop all CO_2 emissions right here and now, the CO_2 we emitted would still take almost a century to reduce to $1/e$ of the emitted amount.
- Mitigation of ozone is financially attractive because the gas damages crops and is a pulmonary irritant.
- People benefit where mitigation is pursued. Crop yields increase, health costs decrease. The beneficial effects of mitigation are local.

Forcing Change From SLCF Reduction



<https://doi.org/10.1073/pnas.1308470110>

Short lived climate forcers SLCP

- Short-lived climate pollutants are powerful climate forcers that remain in the atmosphere for a much shorter period of time than carbon dioxide (CO₂), yet their potential to warm the atmosphere can be many times greater.
- The short-lived climate pollutants black carbon, methane, tropospheric ozone are the most important contributors to the man-made global greenhouse effect after carbon dioxide, responsible for up to 45% of current global warming. If no action to reduce emissions of these pollutants is taken in the coming decades, they are expected to account for as much as half of warming caused by human activity.
- However if we eliminate their emission, where feasible and financially attractive, the effect in terms of reduction of radiative forcing is almost immediate.
- Mitigating SLCP can help us avoid catastrophic climate change.

Black carbon (BC) mitigation

- Black carbon, a climate forcer and air pollutant, is a product of the incomplete combustion of fossil fuels, biofuels and biomass in devices that lack pollution control technology. Most is emitted by solid biofuel use, kerosene lamps, crop residue burning and wildfire.
- The almost complete elimination of Kerosene lights in the past decade has been India's largest contribution to BC mitigation and cut emissions by >40% compared to 2005
- Attempts to shift cooking to cleaner cook stoves aimed at improving health outcomes can result in equally large mitigation over the next decades
- Mitigating open crop residue burning and using the material better (e.g. to produce a biochar soil amendment) can result in mitigation + carbon capture and storage. The alternative is to convert crop residue to compressed biogas and use that to decarbonize the transport sector via CBG fuelled HDVs.
- The third largest source are diesel engines, which can be replaced by EV's (LDV's) and CNG vehicles (HDVs)

Tropospheric ozone mixing ratios

- The transport sector, energy generation and industries are the three largest NO_x generators in India.
- 2-wheelers and 3-wheelers could easily be phased out in favour of EV's which are much cheaper to run ($\frac{1}{6}^{\text{th}}$ of the running cost) and do not depend on charging infrastructure (portable batteries).
- Freight could be shifted to rail if only the construction of the dedicated freight corridors would get finished ($\frac{1}{5}^{\text{th}}$ of the per km cost compared to truck).
- Power generation from renewables is cheaper than from coal
- Reducing the import of oil and coal could reduce the trade deficit.

Methane (CH₄) mitigation

- Rice farming, cattle, sewage and biodegradable waste are the largest easy to mitigate anthropogenic methane sources.
- Innovative rice farming techniques called the system or rice intensification SRI not only increase yields while reducing the water footprint and input cost, they also reduce CH₄ emissions
- Cow dung from large cattle herds can be profitably converted to bio CNG or electricity . Biogas from large scale gober gas contributes around 9% of Germanys electricity supply. This share is negligible in India despite the fact that India has sufficient gober gas potential to substitute all of its gas imports.
- Sewage treatment sludge, crop residue other animal manure, spent wash, press mud and other biodegradable waste can be used in biogas operations as well. Using these will help to reduce landfill methane emissions, clean water bodies often used to dump contaminated water and organic waste and aligns with Swatch Bharat goal.s
- India is pursuing large scale methane mitigation via the Sustainable Alternative Towards Affordable Transportation (SATAT) scheme which aims to set up Compressed Bio-Gas (CBG) production plants from crop residue animal dung and other waste materials. India's gas blending targets mandate increasing Compressed Bio-Gas (CBG) blending in CNG and PNG from 1% in FY2025-26 to 5% by FY2028-29. with 31.4 kilotons CBG sold in 24-25 fiscal against 7,570 kilotons CNG the system currently lags its targets, but so was ethanol blending a decade back.

PRACTICE QUESTIONS

Discuss during lab if you have doubts

Long answer/Short answer

- 1. Define greenhouse gas, global warming potential and radiative forcing**
- 2. Name the most important natural greenhouse gas in the climate system , and list the four most important anthropogenic greenhouse gases . Which of these anthropogenic greenhouse gases are also short lived climate forcers and which ones are long-lived?**
- 3. Explain what the main advantage of aggressive climate change mitigation targeted at short lived climate forcers over aggressive climate change mitigation targeted at long lived species when it comes to stabilizing polar ice caps.**
- 4. What does the terms RCP8.5 stand for ?**
- 5. At present how warm is our climate compared to the pre-industrial reference period?**
- 6. Which climate forcers are called short-lived climate forcers?**

Long answer/Short answer

- 1. Which processes emit the most important ozone precursor NO_x? What can be done to reduce the emissions and how would a society that adopts the necessary changes benefit from the mitigation?**
- 2. Which processes emit most of the BC globally? What has India done to half its BC emission over the past decade and what is India trying to do to accomplish a 90% cut in BC emissions compared to 2005 over the next decade?**
- 3. Where does most of the methane come from and what policies is India currently pursuing to reduce its methane emissions?**